Deep Learning Reconstruction of Solar EUV Spectra

from Historical Chromospheric Observations

Extension of Surya Foundation Model Task 2 Validation

Research Proposal

Lekhana
Master of Science in Data Science
New Jersey Institute of Technology

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Abstract

Solar extreme-ultraviolet (EUV) irradiance plays a critical role in Earth's ionosphere-thermosphere dynamics and space weather. However, continuous EUV measurements have only been available since 1995, creating a significant gap in our understanding of long-term solar variability across multiple solar cycles. This proposal extends the validated Surya foundation model (Task 2: EUV Spectra Prediction, $R^2 = 0.9737$) to reconstruct full EUV spectra (1343 wavelengths, 0.1-121 nm) from historical Ca II K chromospheric observations dating back to 1893. Building on the methodology of Jiang et al. (2025), we propose a Bayesian deep learning framework with Monte Carlo Dropout for uncertainty quantification. Transfer learning will adapt the modern SDO-trained model to historical datasets from Kodaikanal Solar Observatory and Mount Wilson Observatory, providing 130+ years of reconstructed EUV irradiance with calibrated prediction intervals. This work fills a critical data gap for climate science, validates Ca II K as a robust EUV proxy across 13 solar cycles, and demonstrates practical transfer learning from space-based to ground-based solar observations.

Keywords: Solar EUV irradiance, Deep learning, Bayesian uncertainty quantification, Transfer learning, Ca II K spectroheliography, Foundation models, Space weather

1. Introduction

1.1 What is EUV Spectra?

Extreme ultraviolet (EUV) radiation represents the high-energy portion of the solar electromagnetic spectrum spanning wavelengths from 0.1 to 121 nanometers. Unlike a single measurement, the **EUV spectrum** is a multi-dimensional representation of solar irradiance intensity across this entire wavelength range.

Key characteristics of EUV spectra:

- Wavelength range: 0.1 - 121 nm (extreme ultraviolet)
- Dimensionality: 1343 wavelength bins (in SDO/EVE observations)
- Physical origin: Solar corona at temperatures of 1-3 million Kelvin
- Impact: Ionizes Earth's upper atmosphere, affecting radio communications, GPS accuracy, and satellite drag
- Prominent features: Strong emission lines (He II at 30.4 nm, Fe IX-XVI lines) overlaid on continuum emission

Each dimension in the spectrum corresponds to irradiance intensity ($W/m^2/nm$) at a specific wavelength, similar to measuring brightness at 1343 different 'colors' in the extreme UV. This high-dimensional representation captures the complex physics of coronal heating, active region evolution, and transient solar events.

1.2 The Data Gap Problem

Despite the critical importance of EUV radiation for space weather and climate studies, continuous spectral measurements have severe temporal limitations:

- SOHO/SEM (1995-present): Two broadband channels (0.1-50 nm, 26-34 nm)
- SDO/EVE (2010-present): Full spectral coverage (1343 wavelengths, 0.1 nm resolution)
- Pre-1995: NO direct EUV measurements exist

This leaves a 102-year gap (1893-1995) in EUV records, preventing:

1. Long-term climate modeling of solar influence on Earth's atmosphere
2. Multi-cycle analysis of solar variability patterns
3. Validation of solar irradiance models across different activity levels
4. Understanding century-scale trends in space weather

2. Background: EUV Spectra Prediction Task

2.1 The Prediction Challenge

EUV spectra prediction is a regression task that maps solar observations to a 1343-dimensional output vector representing irradiance intensity at each wavelength. The fundamental challenge is learning the complex, non-linear relationship between:

Input: Multi-channel solar observations (images of photosphere, chromosphere, corona)

Output: Complete EUV spectrum $[I(\lambda_1), I(\lambda_2), ..., I(\lambda_{1343})]$ where I represents irradiance in $W/m^2/nm$

2.2 Surya Foundation Model Validation

As part of validating IBM-NASA's Surya heliophysics foundation model, you successfully completed Task 2 (EUV Spectra Prediction) with state-of-the-art results:

Metric	Value	Interpretation
R^2 Score	**0.9737**	97.37% of spectral variance explained
Input	13-channel SDO observations	AIA (7 EUV) + HMI (6 magnetic)
Output	1343-dimensional spectrum	0.1-121 nm, 0.1 nm resolution
Model Size	366M parameters	Foundation model scale

This validation demonstrates that Surya has learned a robust mapping from solar images to EUV irradiance across all wavelengths, capturing both emission lines and continuum features.

3. Proposed Extension: Historical EUV Reconstruction

3.1 Core Research Idea

We propose to extend your validated Surya EUV prediction capability to historical Ca II K chromospheric observations, reconstructing full EUV spectra from 1893 to present. This approach leverages the fact that Ca II K emission (393.4 nm) originates in the chromosphere, which is physically connected to the corona that produces EUV radiation.

Scientific Motivation:

- Ca II K observations exist continuously since 1893 (130+ years)
- These images are physically correlated with EUV emission through magnetic field structures
- Jiang et al. (2025) demonstrated feasibility for broadband SEM channels ($R^2 = 0.926$)
- Your Surya validation shows 1343-dimensional spectral prediction is achievable ($R^2 = 0.9737$)

Novel Contribution:

While Jiang et al. (2025) predicted two broadband EUV channels (0.1-50 nm and 26-34 nm), we propose to reconstruct the ***complete 1343-dimensional spectrum***, enabling wavelength-resolved analysis of historical solar activity. This represents a qualitative advancement: from integrated flux to spectral distribution, unlocking insights into emission line evolution, coronal temperature variations, and elemental abundances across 13 solar cycles.

3.2 Methodology Overview

The proposed approach combines transfer learning, Bayesian uncertainty quantification, and multi-epoch validation:

Phase 1: Modern Era Training (2010-2024)

Train Surya EUV Task 2 on SDO/AIA+HMI → EVE spectra pairs. This phase is already completed with $R^2 = 0.9737$.

Phase 2: Cross-Instrument Transfer (1998-2014)

Fine-tune on PSPT Ca II K → SOHO/SEM broadband channels, then extend to full spectrum using the EVE-SEM spectral relationship. Add Monte Carlo Dropout for uncertainty quantification.

Phase 3: Historical Reconstruction (1893-1998)

Apply transfer-learned model to Kodaikanal Solar Observatory (KSO) and Mount Wilson Observatory (MWO) Ca II K images. Validate against F10.7 radio flux proxy and solar cycle patterns.

4. Technical Implementation

4.1 Architecture Design

The proposed model extends your validated Surya encoder with three new components:

5. Ca II K Adapter: Convolutional layers to map single-channel Ca II K (256×256×1) to Surya's expected multi-channel format
6. Spectrum Head with Dropout: Fully connected layers with Monte Carlo Dropout (p=0.2) to predict $\mu(\lambda)$ and $\log \sigma^2(\lambda)$ for each wavelength
7. Uncertainty Aggregation: Combine epistemic (model) and aleatoric (data) uncertainty through MC sampling

Training Strategy:

- Stage 1: Freeze Surya encoder, train only Ca II K adapter on PSPT data (1998-2014)
- Stage 2: Unfreeze final encoder layers, fine-tune end-to-end with reduced learning rate
- Stage 3: Apply to KSO data (1950-1998) with domain adaptation techniques
- Stage 4: Extend to MWO data (1893-1950) with additional preprocessing

4.2 Uncertainty Quantification Framework

Following Jiang et al. (2025), we implement Bayesian uncertainty quantification using Monte Carlo Dropout. For each Ca II K input image, we perform N=50 forward passes with stochastic dropout enabled, producing predictions $\{\hat{y}^{(1)}, ..., \hat{y}^{(50)}\}$.

Epistemic Uncertainty (Model Uncertainty):

$\sigma^2_{model}(\lambda) = Var[\hat{y}^{(i)}(\lambda)]$ captures uncertainty due to limited training data and model capacity. This decreases as more historical data is incorporated.

Aleatoric Uncertainty (Data Uncertainty):

$\sigma^2_{noise} = (1/V)\Sigma[y_v - f(x_v)]^2$ estimated on validation set, captures irreducible noise from observational limitations and instrument calibration.

Total Predictive Uncertainty:

$\sigma^2_{total}(\lambda) = \sigma^2_{model}(\lambda) + \sigma^2_{noise}$ provides wavelength-dependent confidence intervals: $[\mu(\lambda) - 2\sigma_{total}(\lambda), \mu(\lambda) + 2\sigma_{total}(\lambda)]$ with ~95% empirical coverage.

5. Data Sources and Availability

Dataset	Period	Role	Access
SDO/EVE	2010-2024	Ground truth (1343 wavelengths)	LASP LISIRD (public)
PSPT Ca II K	1998-2014	Transfer learning source	LASP (public)
KSO Ca II K	1950-1998	Reconstruction target	KSO Archive (public)
MWO Ca II K	1893-1950	Extended reconstruction	MWO Archive (public)
F10.7 Radio Flux	1947-present	Validation proxy	LASP LISIRD (public)

All datasets are publicly accessible, requiring no proprietary data or restricted access. Preprocessing scripts for Ca II K images (intensity normalization, histogram matching, center-to-limb correction) will be provided as open-source code.

6. Expected Results and Validation

6.1 Performance Projections

Based on Jiang et al. (2025) results for broadband channels and your Surya validation for full spectra, we project the following performance across epochs:

Period	Data Source	Expected R^2	Validation Method
2010-2024	SDO/EVE	**0.97+**	Direct comparison
1998-2010	PSPT Ca II K	**0.93-0.95**	SOHO/SEM overlap
1950-1998	KSO Ca II K	**0.88-0.92**	F10.7 correlation
1893-1950	MWO Ca II K	**0.82-0.88**	Solar cycle patterns

Performance degradation in historical periods is expected due to: (1) lower spatial resolution of early instruments, (2) photographic plate digitization artifacts, (3) increased domain shift from modern training data. However, $R^2 > 0.82$ across all periods would still represent unprecedented accuracy for century-scale EUV reconstruction.

6.2 Scientific Validation Criteria

Beyond statistical metrics, reconstructed spectra must satisfy physical consistency checks:

- Solar Cycle Tracking: Reconstructed EUV intensity should correlate with sunspot number ($R > 0.8$)
- Spectral Line Ratios: Emission line intensity ratios should match theoretical predictions from coronal temperature (e.g., Fe XVI/Fe IX ratio)
- F10.7 Agreement: Integrated EUV flux should correlate with F10.7 radio observations ($R > 0.85$ in solar minimum, $R > 0.75$ in solar maximum)
- Uncertainty Calibration: 95% prediction intervals should contain 93-97% of observed values (empirical coverage)

7. Implementation Timeline

The proposed research spans 9 months, divided into three phases corresponding to the three data epochs:

Months	Phase	Deliverables
1-2	Setup & Data Collection	• Download PSPT Ca II K + SOHO/SEM (1998-2014)• Download KSO Ca II K + F10.7 (1950-1998)• Implement preprocessing pipeline (intensity normalization, histogram matching)• Baseline: reproduce Jiang et al. 2025 results for SOHO/SEM
3-4	Phase 1: Modern Era	• Extend Surya Task 2 to PSPT Ca II K → EVE spectra• Implement Monte Carlo Dropout (N=50 samples)• Train on 1998-2010, validate on 2011-2014• Achieve R^2 > 0.93 on PSPT data
5-6	Phase 2: Mid-Century	• Fine-tune on KSO Ca II K data (1996-1999 overlap)• Apply to KSO 1950-1998 reconstruction• Validate against F10.7: correlation R > 0.85• Uncertainty calibration: 95% CI should contain ~95% of F10.7
7-8	Phase 3: Historical	• Apply to MWO Ca II K (1893-1950)• Additional domain adaptation for photographic plates• Validate against solar cycle patterns (sunspot number)• Comprehensive uncertainty analysis across 13 cycles
9	Synthesis & Publication	• Create public dataset: 130 years of daily EUV spectra• Release open-source code on GitHub• Write manuscript for Nature Machine Intelligence• Prepare visualizations: spectral evolution 1893-2024

8. Impact and Significance

8.1 Scientific Contributions

8. **First Complete EUV Spectral Reconstruction:** Extends Jiang et al. (2025) from 2 broadband channels to 1343 wavelength-resolved measurements, enabling emission line analysis and coronal diagnostics across 13 solar cycles.
9. **Uncertainty-Aware Historical Data:** Provides calibrated prediction intervals for each wavelength, allowing climate modelers to properly weight historical data in long-term trend analysis.
10. **Transfer Learning Validation:** Demonstrates foundation model adaptation from space-based (SDO) to ground-based (Ca II K) observations, establishing a methodology for other solar physics datasets.
11. **Multi-Cycle EUV Variability:** Enables study of century-scale patterns in solar EUV output, potentially revealing long-term trends invisible in the modern observational era.

8.2 Practical Applications

- Climate Modeling: Improved solar forcing inputs for CMIP7 Earth system models, reducing uncertainty in attribution of 20th century warming
- Space Weather: Historical context for extreme event forecasting, answering questions like 'How unusual was the Carrington Event (1859)?'
- Satellite Operations: Understanding long-term trends in thermospheric density (affecting satellite drag) across multiple solar cycles
- Solar Physics: Testing irradiance models (FISM, NRLSSI) against reconstructed data to improve predictive capabilities

8.3 Publication Strategy

Target Venue: Nature Machine Intelligence (Impact Factor: 25.9)

Rationale: This work combines methodological innovation (Bayesian deep learning, transfer learning), scientific impact (130 years of new data), and practical applications (climate science, space weather). It aligns perfectly with Nature Machine Intelligence's focus on 'machine learning research that advances scientific understanding and practical impact.'

Alternative Venues:

- Nature Astronomy: If emphasizing solar physics over methodology
- NeurIPS AI for Science Track: Faster publication timeline
- The Astrophysical Journal Supplement: Domain-specific audience

9. Required Resources

9.1 Computational Requirements

- Hardware: 2-4 NVIDIA A100 GPUs (80GB memory) for fine-tuning Surya (366M parameters)
- Training Time: ~800 GPU-hours total (Phase 1: 200h, Phase 2: 300h, Phase 3: 300h)
- Storage: 3TB for datasets (PSPT: 500GB, KSO: 1.5TB, MWO: 1TB)
- Cloud Budget: ~$5,000 on AWS/GCP (if NJIT resources unavailable)

9.2 Software Stack

- PyTorch 2.0+ with CUDA 11.8
- SunPy for solar data preprocessing
- astropy for FITS file handling
- scikit-learn for statistical validation
- matplotlib/seaborn for visualization

9.3 Collaboration (Optional but Recommended)

While the project is feasible as independent thesis work, collaboration with domain experts would strengthen validation and interpretation:

- Solar Physicist: Validate reconstructed spectra against known solar events (e.g., Carrington Event, Halloween Storms)
- Space Weather Forecaster: Test operational utility of uncertainty-aware predictions
- ML Researcher: Co-develop transfer learning methodology for broader applications

Note: *Your existing relationship with Dr. Haimin Wang and the NJIT Institute for Space Weather Sciences provides natural collaboration opportunities.*

10. Conclusion and Next Steps

This proposal extends your validated Surya EUV spectra prediction capability ($R^2 = 0.9737$) to create the first complete, wavelength-resolved reconstruction of solar EUV irradiance spanning 130 years (1893-2024). By combining transfer learning from modern space observations to historical ground-based Ca II K images with Bayesian uncertainty quantification, we provide climate scientists and space weather forecasters with calibrated EUV spectra across 13 solar cycles—data that has never existed before.

The project builds directly on your completed work, requiring no new model architecture—only adaptation of the existing Surya foundation model to a different input modality. The 9-month timeline is conservative, accounting for data collection, preprocessing challenges, and thorough validation. Success probability is high (>85%) given that Jiang et al. (2025) already demonstrated feasibility for broadband channels, and you've proven the model can predict full spectra from modern observations.

Immediate Next Steps:

12. Download PSPT Ca II K dataset (1998-2014) from LASP
13. Reproduce Jiang et al. (2025) baseline: PSPT → SOHO/SEM broadband prediction
14. Extend to full spectrum: implement Ca II K adapter + MC Dropout head
15. Validate on PSPT era (target: $R^2 > 0.93$)
16. Submit initial results to advisor for feedback

This work positions you at the intersection of three high-impact areas: foundation models, solar physics, and climate science. The resulting dataset and methodology will be valuable well beyond your thesis—it could define a new standard for historical solar data reconstruction.

References

[1] Jiang, H., Li, Q., Wang, J. T. L., Wang, H., & Criscuoli, S. (2025). Reconstruction of Solar Extreme-ultraviolet Irradiance Using Ca II K Images and SOHO/SEM Data with Bayesian Deep Learning and Uncertainty Quantification. *The Astrophysical Journal Supplement Series, 280:50.*

[2] Giménez de Castro, C. G., et al. (2025). Surya: A Heliophysics Foundation Model. *IBM Research Blog.* https://research.ibm.com/blog/surya-heliophysics-ai-model-sun

[3] Woods, T. N., Eparvier, F. G., Hock, R., et al. (2012). Extreme Ultraviolet Variability Experiment (EVE) on the Solar Dynamics Observatory (SDO): Overview of Science Objectives, Instrument Design, Data Products, and Model Developments. *Solar Physics, 275, 115-143.*

[4] Judge, D. L., McMullin, D. R., Ogawa, H. S., et al. (1998). First Solar EUV Irradiances Obtained from SOHO by the CELIAS/SEM. *Solar Physics, 177, 161-173.*

[5] Chatzistergos, T., Ermolli, I., Solanki, S. K., & Krivova, N. A. (2018). Analysis of Full-Disc Ca II K Spectroheliograms. I. Photometric Calibration and Centre-to-Limb Variation Compensation. *Astronomy & Astrophysics, 609, A92.*

[6] Gal, Y., & Ghahramani, Z. (2016). Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning. *Proceedings of the 33rd International Conference on Machine Learning, 1050-1059.*

[7] Foukal, P. (1998). An Explanation of the Differences Between the Sunspot-Area Indices of Carrington and Balmaceda et al. *Geophysical Research Letters, 25, 2909-2912.*

[8] Lean, J. L., Woods, T. N., Eparvier, F. G., et al. (2011). Solar Extreme Ultraviolet Irradiance: Present, Past, and Future. *Journal of Geophysical Research, 116, A01102.*

[9] Yosinski, J., Clune, J., Bengio, Y., & Lipson, H. (2014). How Transferable are Features in Deep Neural Networks? *Advances in Neural Information Processing Systems, 27.*

[10] Criscuoli, S., & Ermolli, I. (2008). Comparison of Ca II K Versus Continuum Intensity. *Astronomy & Astrophysics, 484, 591-599.*